Lecture 7 – Equivalence of Regular Expressions and Finite Automata

COSE215: Theory of Computation

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PLRG

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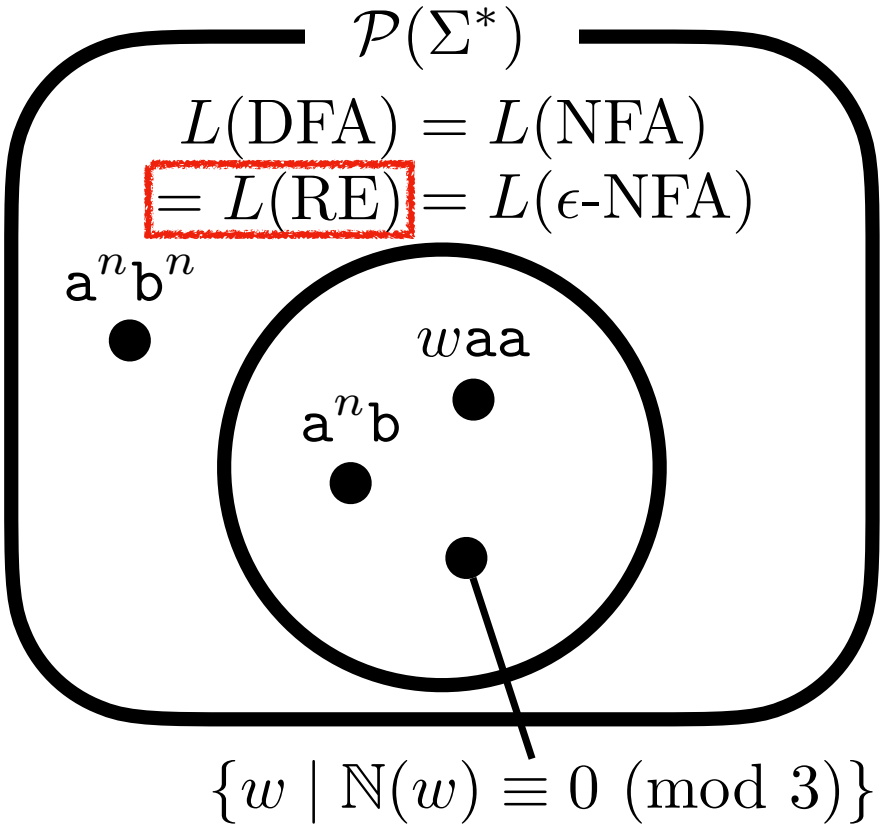
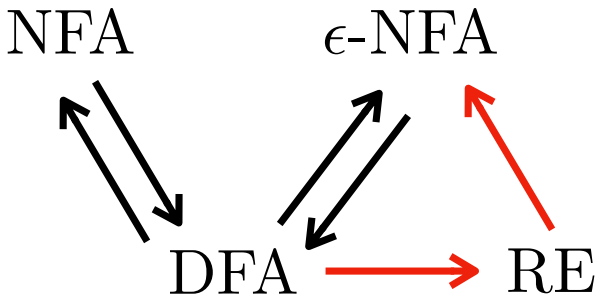
Recall

- Regular Expressions
  - Operations in languages
  - Definition
  - Precedence order
  - Language of regular expressions
  - Extended regular expressions
  - Examples
- Regular Expressions in Practice

Equivalence of REs and FA





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2. DFA to Regular Expressions
   - Inductive Construction of Regular Expressions
   - State Elimination Method

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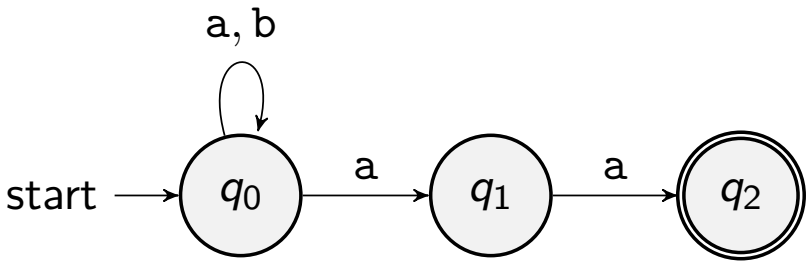
Regular Expressions to ϵ-NFA

Theorem (Regular Expressions to ϵ-NFA)

For a given regular expression R, $\exists$ ϵ-NFA N^ϵ. $L(R) = L(N^\epsilon)$.

`(a|b)*aa` ➡

ϵ-NFA N^ϵ

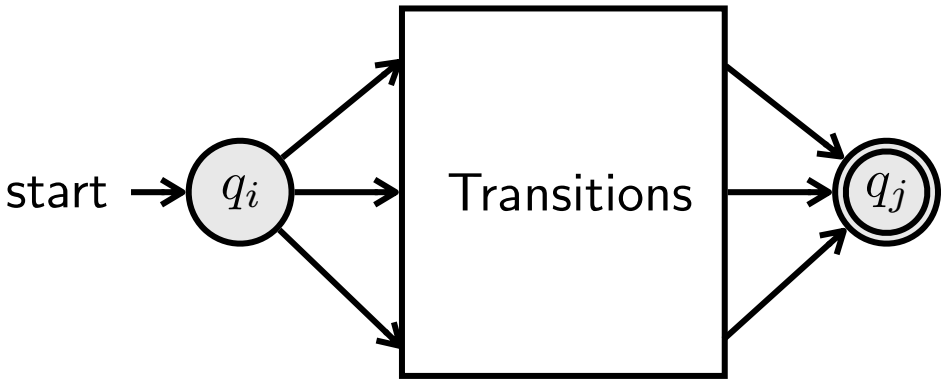


Regular Expressions to ϵ-NFA

For a given regular expression R and an integer i, we will construct an ϵ-NFA $N^\epsilon = (Q, \Sigma, \delta, q_i, F)$ that accepts the language of R.

It satisfies the following properties:

- States are q_i, q_{i+1}, $\cdots$, and q_j ($Q = \{q_k \mid i \leq k \leq j\}$)
- The last state is the unique final state ($F = \{q_j\}$)
- No transition to the initial state ($\forall q \in Q.\ \forall a \in \Sigma \cup \{\epsilon\}.\ q_i \notin \delta(q, a)$)
- No transition from the final state ($\forall a \in \Sigma \cup \{\epsilon\}.\ \delta(q_j, a) = \varnothing$)

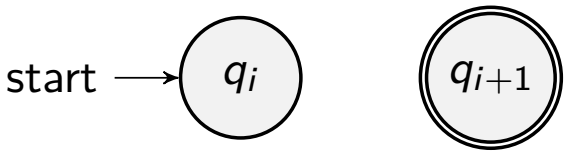


ϵ-NFA for (R, i)

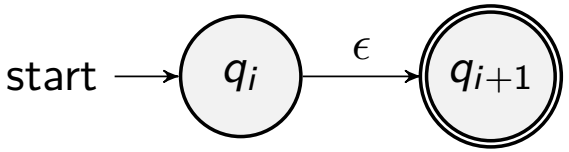
Regular Expressions to ϵ-NFA

For a given regular expression R and an integer i, the ϵ-NFA for (R, i) is:

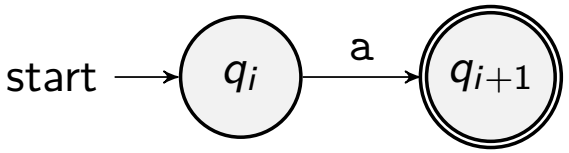
- $R = \varnothing$:



- $R = \epsilon$:

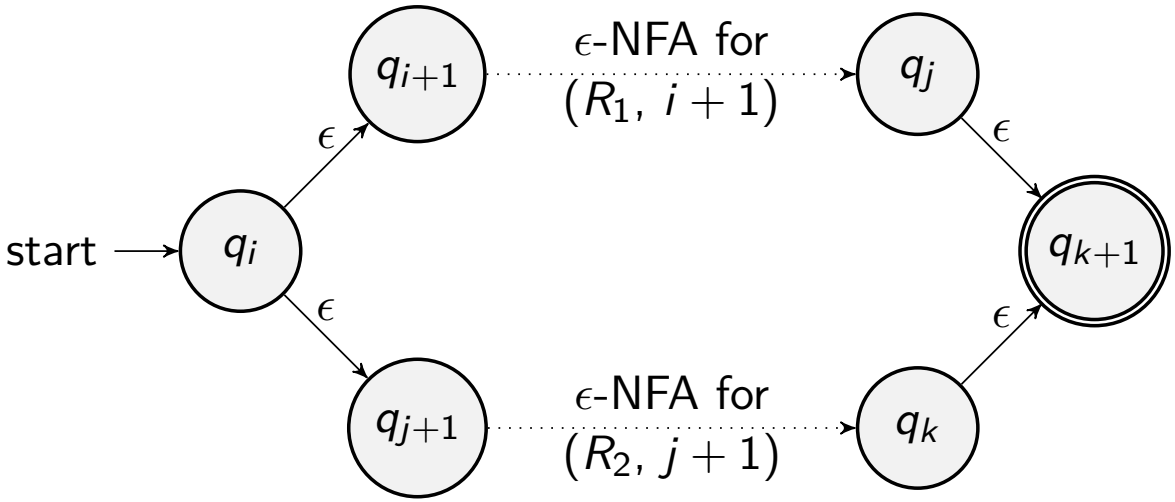


- $R = \mathtt{a}$:



Regular Expressions to ϵ-NFA

- $R = R_1 \mid R_2$:



Regular Expressions to ϵ-NFA

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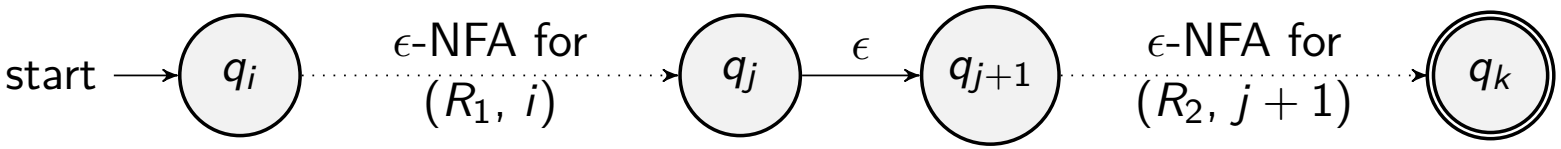
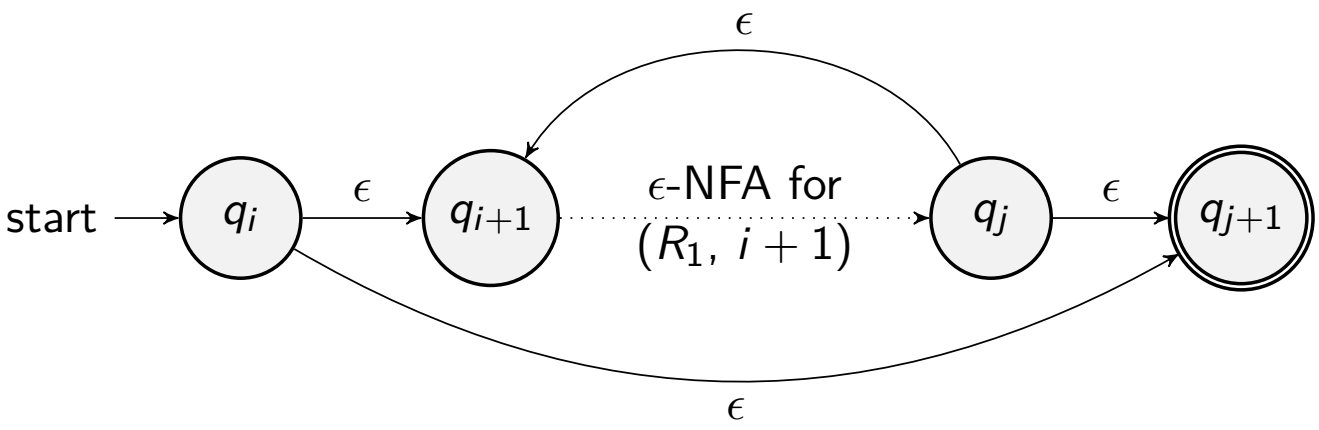


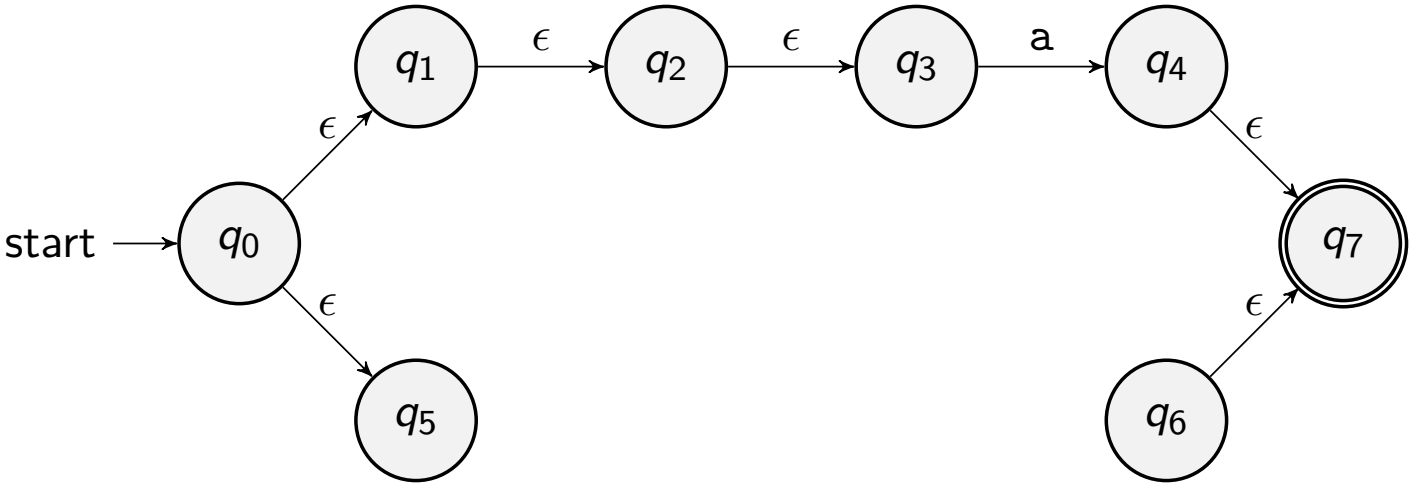
Regular Expressions to ϵ-NFA

- $R = R_1^*$:



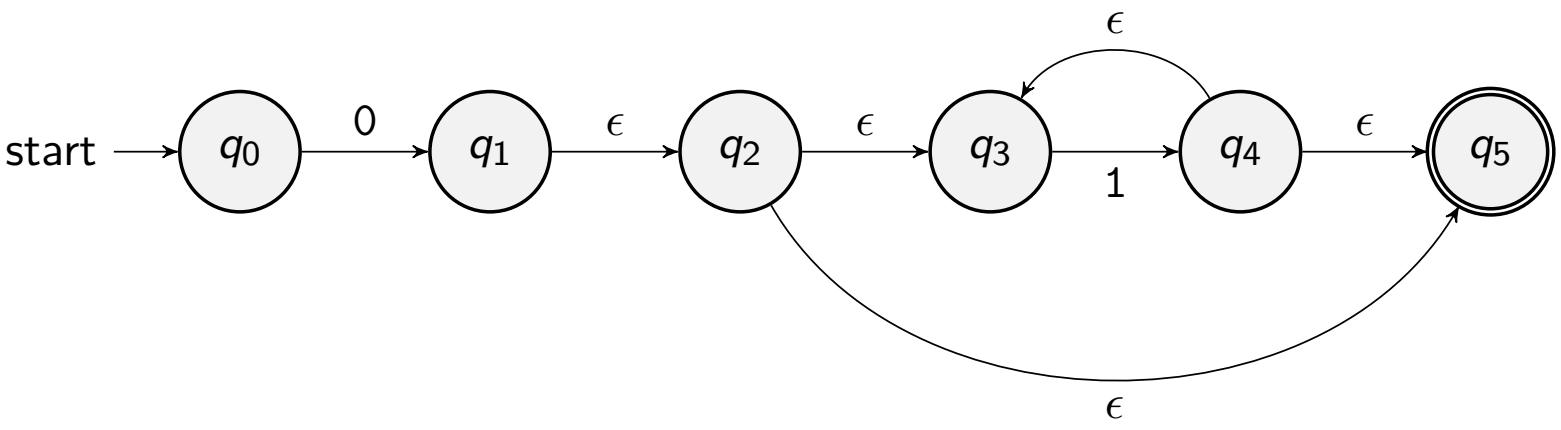
Regular Expressions to ϵ-NFA – Examples

- $R = \epsilon\mathtt{a} \mid \varnothing$



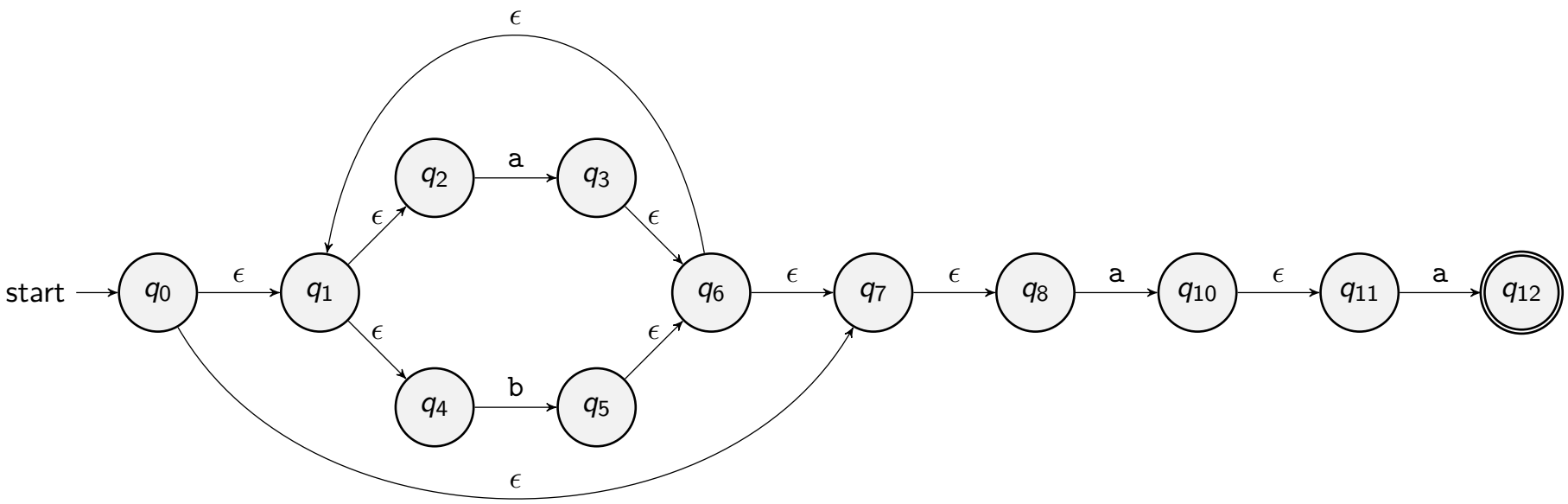
Regular Expressions to ϵ-NFA – Examples

- $R = 01^*$



Regular Expressions to ϵ-NFA – Examples

- $R = (\mathtt{a}|\mathtt{b})^*\mathtt{aa}$



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DFA to Regular Expressions

Theorem (DFA to Regular Expressions)

For a given DFA $D = (Q, \Sigma, \delta, q_1, F)$, $\exists$ RE R. $L(D) = L(R)$.

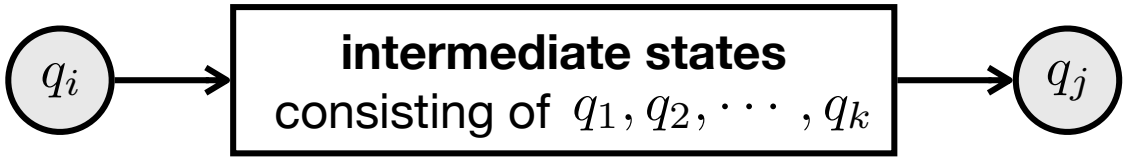
We will learn two different way to convert a DFA to a regular expression.

1. **Inductive Construction** of **Regular Expressions** for paths in a DFA with bounded intermediate states where $Q = \{q_1, q_2, \cdots, q_n\}$.

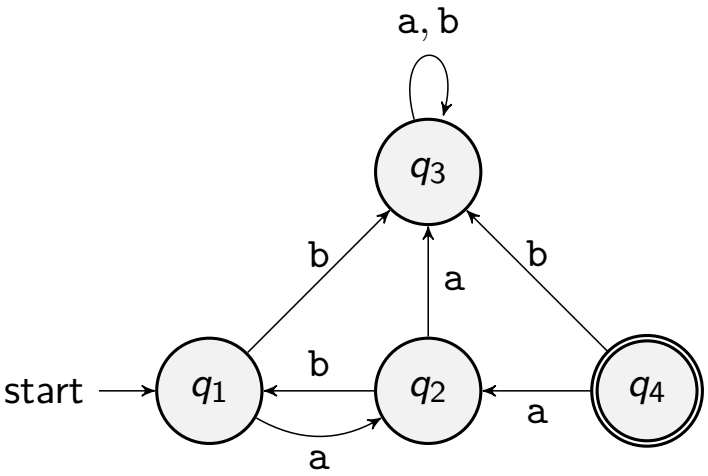
2. **State Elimination Method** in an extended DFA using regular expressions as labels.

Inductive Construction of Regular Expressions

Let $R_{i,j}^{(k)}$ be the **regular expression** that accepts the **paths** from q_i to q_j whose indices of the **intermediate** states are **bounded** by k.



For example, $R_{1,3}^{(2)}$ is the regular expression that accepts the paths from q_1 to q_3 whose intermediate states are q_1 and q_2.



$L(R_{1,3}^{(2)}) \ni$ b a a a b a a

$\not\ni$ a b a a a b

Inductive Construction of Regular Expressions

We can **inductively construct** regular expressions $R_{i,j}^{(k)}$ for all combination of i, j, and k (induction on k).

- **(Basis Case)** $k = 0$

  It means that **no intermediate states** in the path.

  - If $i \neq j$ (source and destination states are different),

    $$R_{i,j}^{(0)} = \mathtt{a}_1 \,|\, \mathtt{a}_2 \,|\, \cdots \,|\, \mathtt{a}_m$$

    where $q_i \xrightarrow{\mathtt{a}_1} q_j, q_i \xrightarrow{\mathtt{a}_2} q_j, \cdots, q_i \xrightarrow{\mathtt{a}_m} q_j$ are transitions in D.

  - If $i = j$ (source and destination states are same),

    $$R_{i,j}^{(0)} = R_{i,i}^{(0)} = \epsilon \,|\, \mathtt{a}_1 \,|\, \mathtt{a}_2 \,|\, \cdots \,|\, \mathtt{a}_m$$

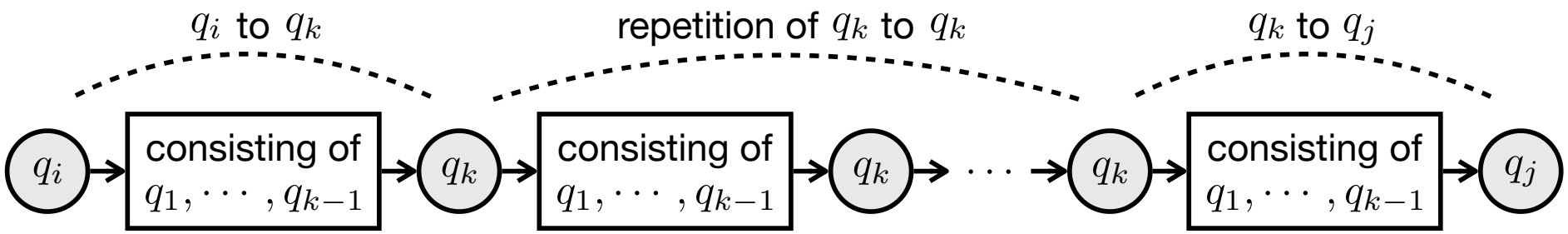
    where $q_i \xrightarrow{\mathtt{a}_1} q_i, q_i \xrightarrow{\mathtt{a}_2} q_i, \cdots, q_i \xrightarrow{\mathtt{a}_m} q_i$ are transitions in D.

Inductive Construction of Regular Expressions

- **(Induction Case)** $R_{i,j}^{(k-1)}$ are given for all i and j.

$$R_{i,j}^{(k)} = R_{i,j}^{(k-1)} \mid R_{i,k}^{(k-1)} (R_{k,k}^{(k-1)})^* R_{k,j}^{(k-1)}$$

  - $R_{i,j}^{(k-1)}$: paths from q_i to q_j **NOT** containing q_k as intermediate states.
  - $R_{i,k}^{(k-1)} (R_{k,k}^{(k-1)})^* R_{k,j}^{(k-1)}$: paths from q_i to q_j containing q_k at least once as intermediate states.



Inductive Construction of Regular Expressions

Consider the following DFA:

$$D = (Q, \Sigma, \delta, q_1, F)$$

where $Q = \{q_1, q_2, \cdots, q_n\}$ and $F = \{q_{f_1}, q_{f_2}, \cdots, q_{f_m}\}$.

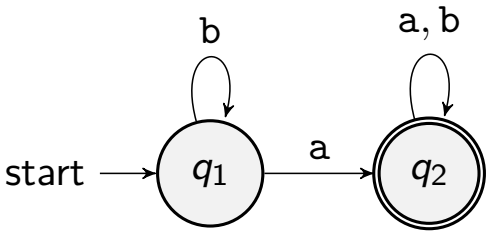
Then, using the regular expressions $R_{i,j}^{(k)}$ with bounded intermediate states, we can construct the regular expression R that accepts the language of the DFA D as follows:

$$R = R_{1,f_1}^{(n)} \mid R_{1,f_2}^{(n)} \mid \cdots \mid R_{1,f_m}^{(n)}$$

The regular expression R accepts all the paths from the **initial state** q_1 to one of the **final states** q_{f_1}, q_{f_2}, $\cdots$, q_{f_m} but **no bound** on the intermediate states (because $k = n$).

Examples

Let's construct the regular expressions $R_{i,j}^{(k)}$ for the following DFA:



When $k = 0$, we have:

- $R_{1,1}^{(0)} = \epsilon \mid \mathtt{b}$
- $R_{1,2}^{(0)} = \mathtt{a}$
- $R_{2,1}^{(0)} = \varnothing$
- $R_{2,2}^{(0)} = \epsilon \mid \mathtt{a} \mid \mathtt{b}$

Examples

When $k = 0$, we have:

- $R^{(0)}_{1,1} = \epsilon \mid \mathtt{b}$
- $R^{(0)}_{1,2} = \mathtt{a}$
- $R^{(0)}_{2,1} = \varnothing$
- $R^{(0)}_{2,2} = \epsilon \mid \mathtt{a} \mid \mathtt{b}$

When $k = 1$, we have:

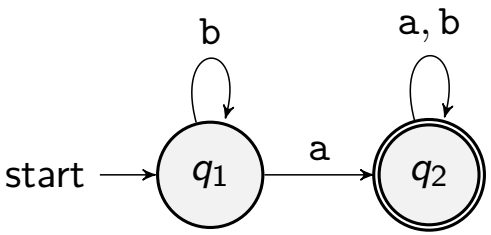
- $R^{(1)}_{1,1} = R^{(0)}_{1,1} \mid R^{(0)}_{1,1}(R^{(0)}_{1,1})^* R^{(0)}_{1,1} = (R^{(0)}_{1,1})^+ = (\epsilon \mid \mathtt{b})^+ = \mathtt{b}^*$
- $R^{(1)}_{1,2} = R^{(0)}_{1,2} \mid R^{(0)}_{1,1}(R^{(0)}_{1,1})^* R^{(0)}_{1,2} = (R^{(0)}_{1,1})^* R^{(0)}_{1,2} = (\epsilon \mid \mathtt{b})^* \mathtt{a} = \mathtt{b}^* \mathtt{a}$
- $R^{(1)}_{2,1} = R^{(0)}_{2,1} \mid R^{(0)}_{2,1}(R^{(0)}_{1,1})^* R^{(0)}_{1,1} = \varnothing \mid \varnothing = \varnothing$
- $R^{(1)}_{2,2} = R^{(0)}_{2,2} \mid R^{(0)}_{2,1}(R^{(0)}_{1,1})^* R^{(0)}_{1,2} = R^{(0)}_{2,2} \mid \varnothing = R^{(0)}_{2,2} = \epsilon \mid \mathtt{a} \mid \mathtt{b}$

Examples

When $k = 1$, we have:

- $R^{(1)}_{1,1} = \mathtt{b}^*$
- $R^{(1)}_{1,2} = \mathtt{b}^*\mathtt{a}$
- $R^{(1)}_{2,1} = \varnothing$
- $R^{(1)}_{2,2} = \epsilon \mid \mathtt{a} \mid \mathtt{b}$

Let's focus on the regular expression for the language of the DFA.



We need to compute $R^{(2)}_{1,2}$ using the results when $k = 1$:

- $$\begin{aligned} R^{(2)}_{1,2} &= R^{(1)}_{1,2} \mid R^{(1)}_{1,2}(R^{(1)}_{2,2})^* R^{(1)}_{2,2} = R^{(1)}_{1,2}(R^{(1)}_{2,2})^* \\ &= \mathtt{b}^*\mathtt{a}(\epsilon \mid \mathtt{a} \mid \mathtt{b})^* \\ &= \mathtt{b}^*\mathtt{a}(\mathtt{a} \mid \mathtt{b})^* \qquad \text{(the regular expression for the above DFA)} \end{aligned}$$

State Elimination Method

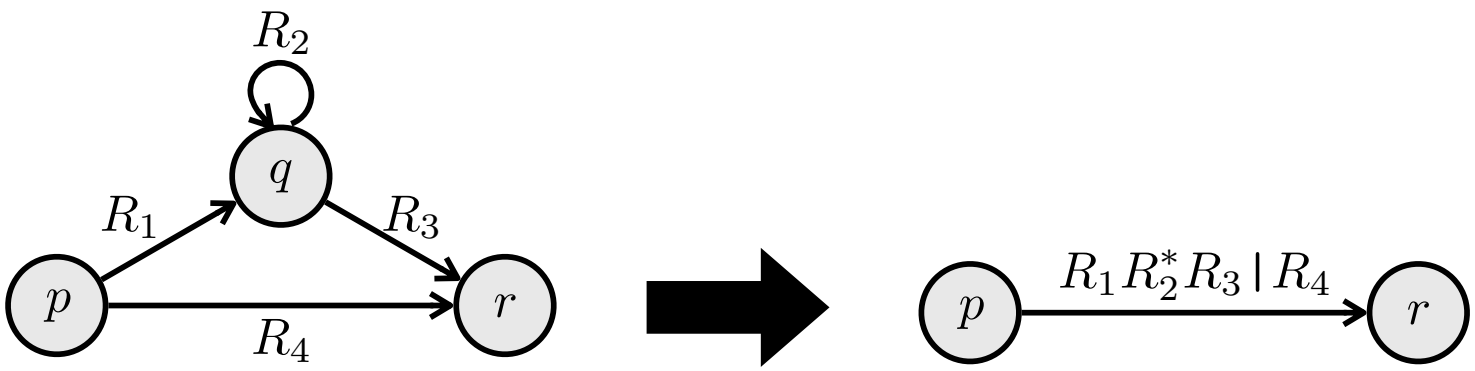
- There is another way to convert a DFA to a regular expression.
- It is **more intuitive** and easier to understand but **not easy to implement**.
- The idea is to **eliminate** the **states** of the DFA one by one and **construct the regular expressions**.
- We will assign constructed **regular expressions** instead of symbols as **labels** on the transitions between the states in the DFA.

State Elimination Method

We can convert a DFA to a regular expression using the following steps for **each final state** $q_f \in F$:

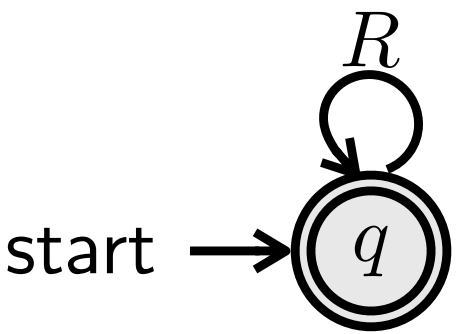
1. **Merge** symbols $a_1, a_2, \cdots, a_m$ on the transition from q_i to q_j into a single regular expression $a_1 \mid a_2 \mid \cdots \mid a_m$.

2. **Eliminate** a state q that is **not** the **initial** state or the **target final** state using the following mechanism:

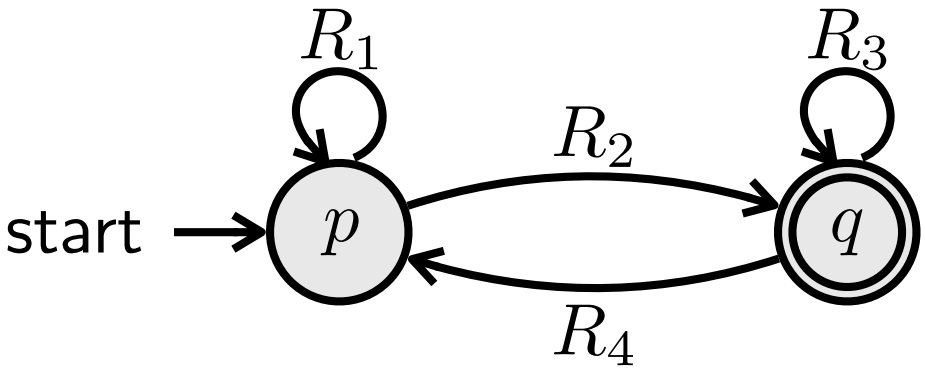


State Elimination Method

3. **Construct** regular expressions for the remaining one or two states using the regular expressions on the transitions between the states.



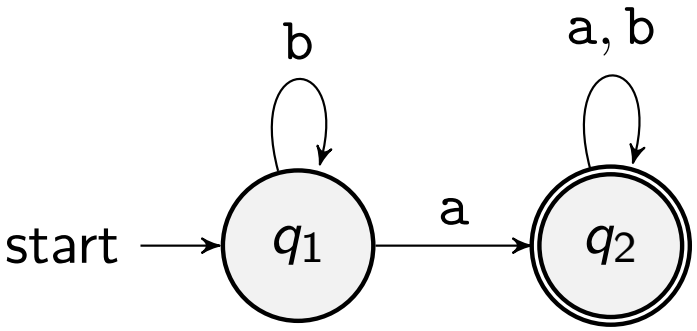
R^*



$(R_1 \mid R_2 R_3^* R_4)^* R_2 R_3^*$

Example 1

Let's convert the following DFA to a regular expression using the **state elimination** method:



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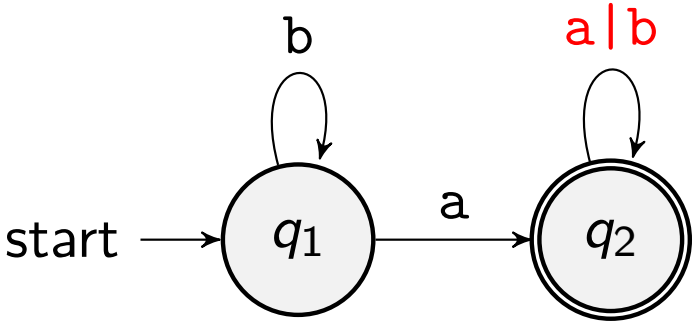
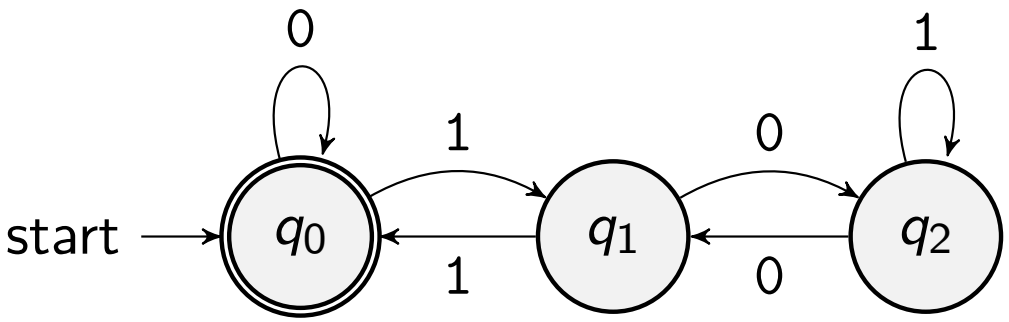


The regular expression for the above DFA is:

$$\mathtt{b}^*\mathtt{a}(\mathtt{a}|\mathtt{b})^*$$

Example 2

The following DFA accepts $L = \{w \in \{0,1\}^* \mid \mathbb{N}(w) \equiv 0 \pmod 3\}$ where $\mathbb{N}(w)$ is the natural number represented by w in binary:



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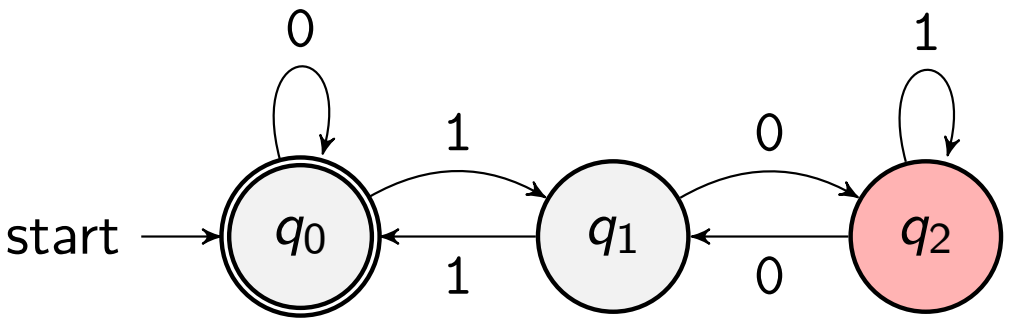

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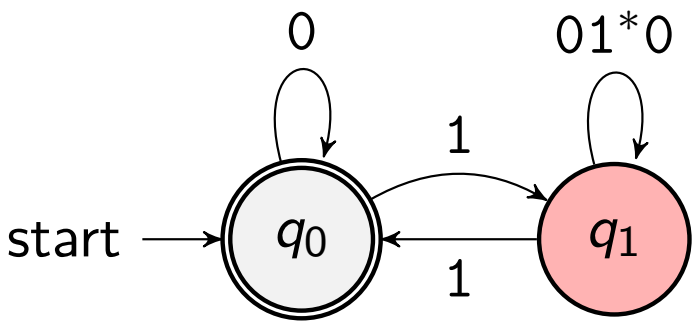

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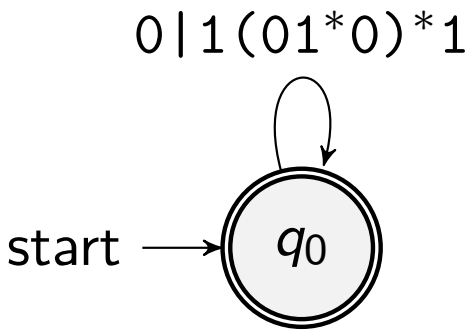
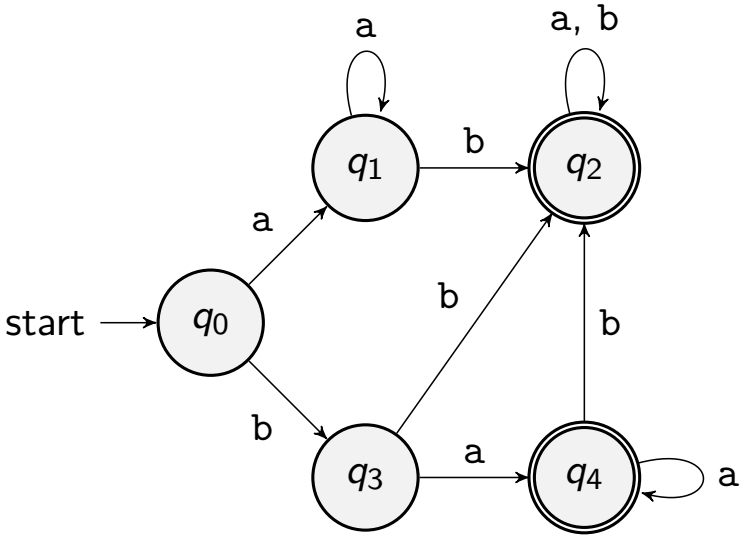


Then, the regular expression for the above DFA is:

$$(0|1(01^*0)^*1)^*$$

Example 3

Let's convert the following DFA to a regular expression using the **state elimination** method:

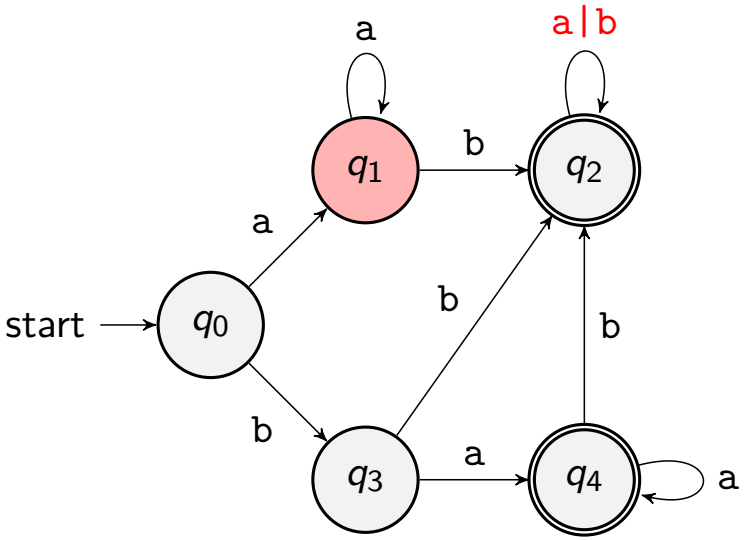


We need to consider two final states q_2 and q_4.

Let's start by eliminating non-initial and non-final states.

Example 3

Let's convert the following DFA to a regular expression using the **state elimination** method:



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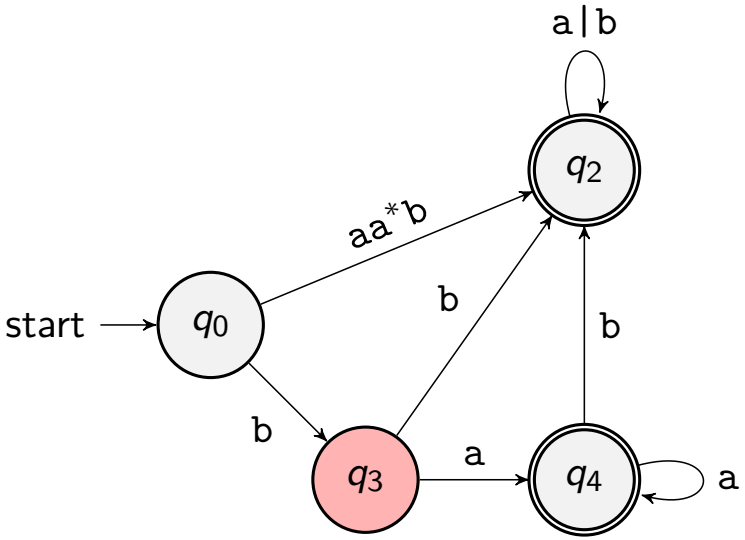

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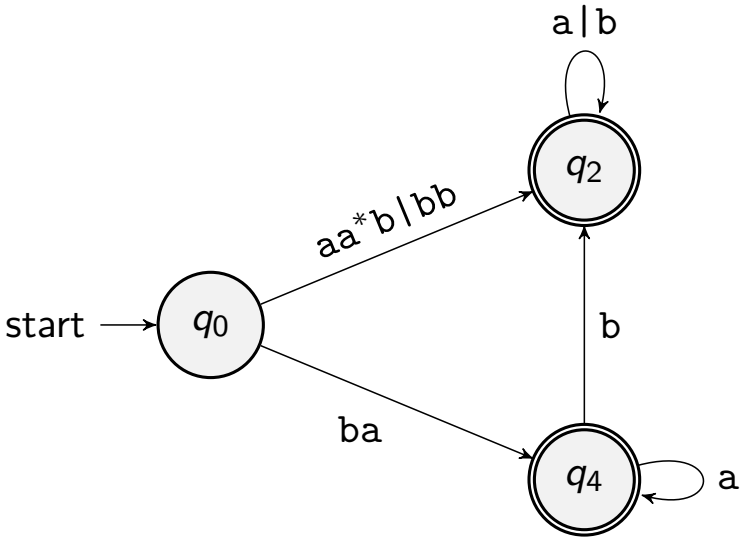
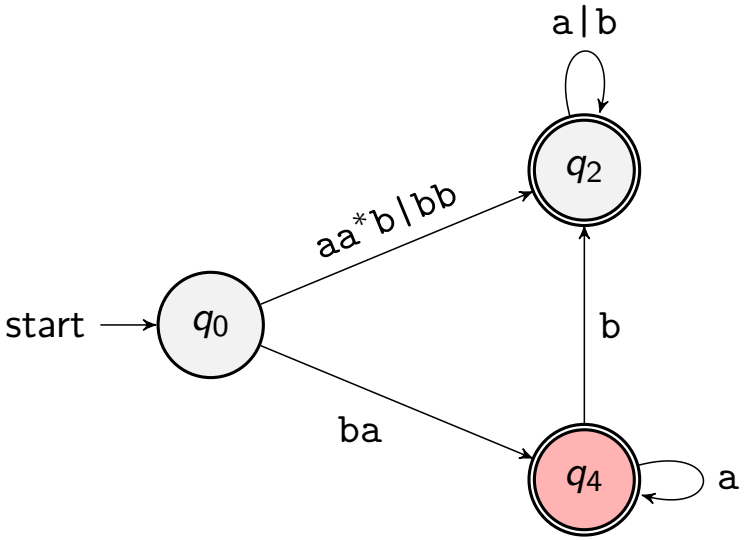


Now, we know that the regular expression for the **final state** q_4 is:

$$\texttt{baa}^*$$

Example 3

Let's convert the following DFA to a regular expression using the **state elimination** method:



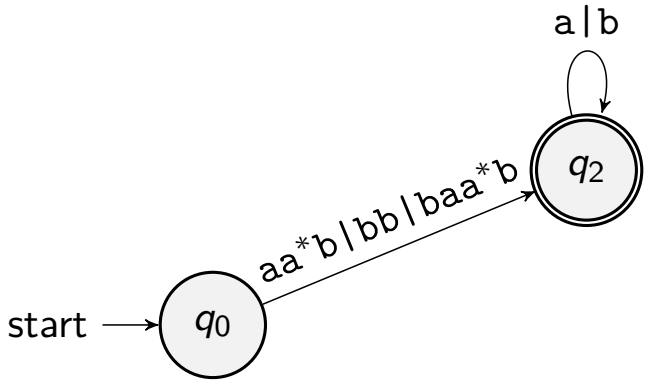
Now, we know that the regular expression for the **final state** q_4 is:

$$\texttt{baa}^*$$

Let's keep eliminating q_4 to know the regular expression for q_2.

Example 3

Let's convert the following DFA to a regular expression using the **state elimination** method:

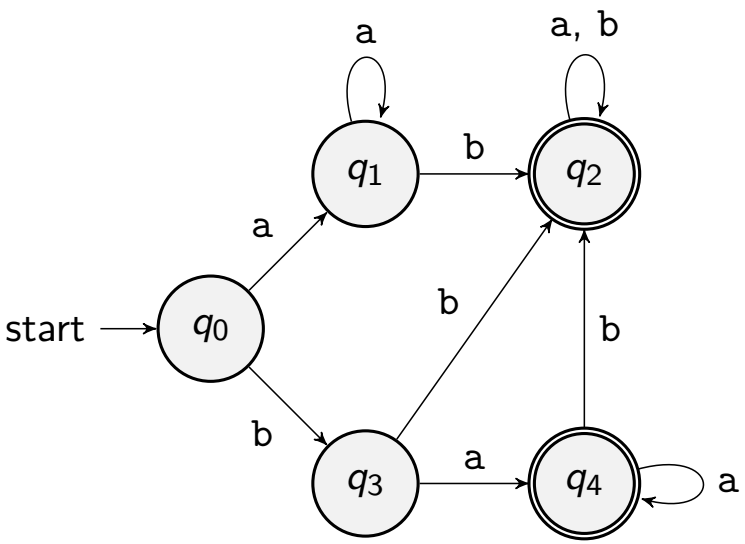


Now, we know that the regular expression for the **final state** q_2 is:

$$(\mathtt{aa}^*\mathtt{b}|\mathtt{bb}|\mathtt{baa}^*\mathtt{b})(\mathtt{a}|\mathtt{b})^*$$

Example 3

Let's convert the following DFA to a regular expression using the **state elimination** method:

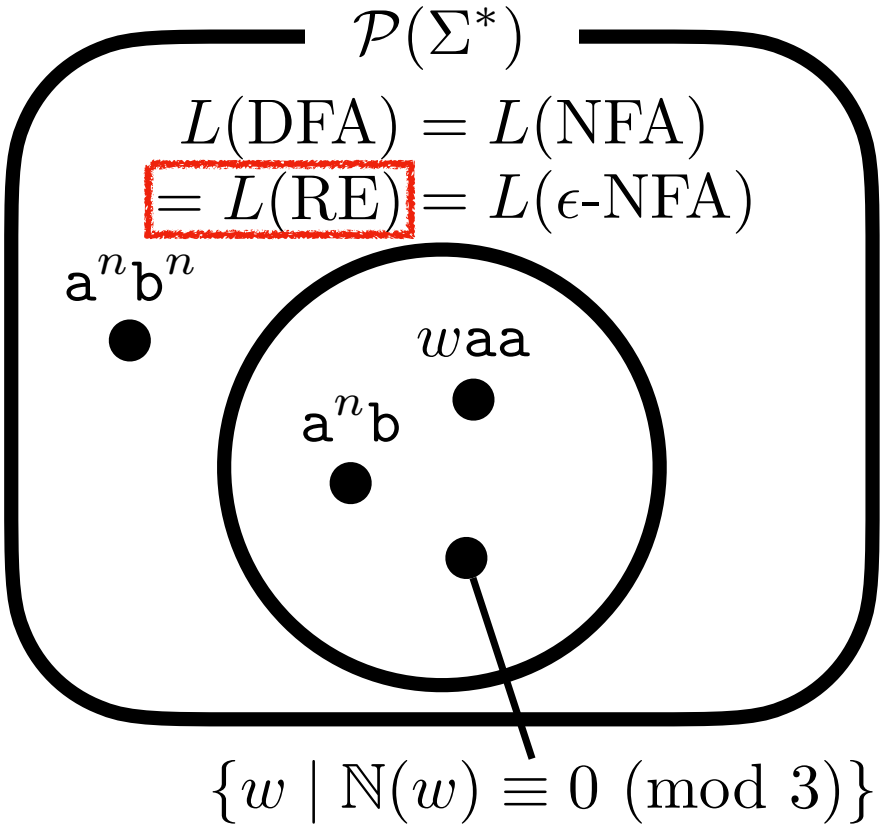


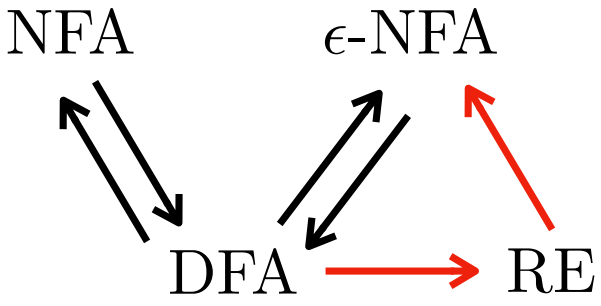
Finally, we have the regular expression for the above DFA:

$$(aa^*b|bb|baa^*b)(a|b)^*|baa^*$$

Note that $(aa^*b|bb|baa^*b)(a|b)^*$ is for q_2 and baa^* is for q_4.

Summary


$\mathcal{P}(\Sigma^*)$
$L(\text{DFA}) = L(\text{NFA})$
$= L(\text{RE}) = L(\epsilon\text{-NFA})$
$\mathtt{a}^n\mathtt{b}^n$
$w\mathtt{aa}$
$\mathtt{a}^n\mathtt{b}$
$\{w \mid \mathbb{N}(w) \equiv 0 \pmod 3\}$


NFA
ϵ-NFA
DFA
RE

Exercise #2

- Please see this document for the exercise.

https://github.com/ku-plrg-classroom/docs/tree/main/cose215/equiv-re-fa

- Please implement the following functions in `Implementation.scala`.
  - `reToENFA` for the conversion from REs to ϵ-NFAs.
  - `dfaToRE` for the conversion from DFAs to REs.
  - `enfaToDFA` for the conversion from ϵ-NFAs to DFAs.
- It is just an exercise, and you **don't need to submit** anything.

Next Lecture

- Closure Properties of Regular Languages

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